Abigail Woodring

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Education

Ph.D. in Electrical Engineering, North Carolina State University — Expected May 2029
Master of Science, Electrical Engineering, North Carolina State University — May 2026, GPA 4.0/4.0
Bachelor of Science, Electrical Engineering, North Carolina State University — May 2025, GPA 4.0/4.0

Publications

[1] **Abigail Woodring**, Adrian Chan, Rana Muhammad Shahroz Khan, Sukwon Yun, Chau-Wai Wong, and Tianlong Chen, "The blessing of dimensionality: How near-orthogonality in high-dimensional spaces explains temporal portability," under review for *Journal of Machine Learning Research (JMLR)*. [arXiv] [Source Code]

[2] Adrian Chan, **Abigail Woodring**, Rana Shahroz, Sukwon Yun, Chau-Wai Wong, Tianlong Chen, "How long do model patches last? A temporal perspective on PortLLM," *Preprint*. [OpenReview]

Awards

Silver Reviewer Award, International Conference on Machine Learning (ICML) — May 2026
Goodnight Doctoral Fellowship, NC State University — 2025 - Present
Provost Doctoral Fellowship, NC State University — 2025 - 26
Scholarly Achievement Award, NC State, Department of Electrical and Computer Engineering — May 2025
Faculty Senior Scholar Award, NC State, Department of Electrical and Computer Engineering — May 2025

Research Experiences

Machine Learning with Sensitive Data Using Differential Privacy — Sept. 2025 - Present

- Uncovered failure in standard private learning algorithm via mathematical analysis of gradient descent steps
- Developing algorithm to alleviate this failure by focusing on the direction of the perturbed gradient descent steps

Joint Experimental and Theoretical Study of LLM's Portability (led to first-authored paper [1]) — Feb. - Jul. 2026

- Conducted large-scale experiments on ten discrete pretraining phases for Mistral, Gemma, and Qwen models and fine-tuned and evaluated on WinoGrande, BoolQ, ARC-Easy, and ARC-Challenge datasets to unveil the temporal portability of LoRA patches on evolved LLMs
- Devised two mathematical analyses on LLM parameters and LoRA patches to discover the root cause of experimentally observed temporal portability
- Revealed that near-orthogonality of high-dimensional vectors is fundamental to both analyses, with potential to be used as a general tool to compare LLM patching strategies

Statistical Algorithms to Forecast LLM's Portability (led to second-authored paper [2]) — Feb. 2025 - Jan. 2026

- Derived hypothesis test to forecast performance and showed an AUC above 0.8 for 5 out of 8 test scenarios
- Developed statistical estimation algorithm to forecast performance and showed statistical significance ($p\text{-value}<0.05$) for 65 out of 72 test cases
- Contributed 4 out of 9 pages to the main text of the paper and 10 out of 14 pages to the paper's appendices

Work Experiences

RF Applications Intern, Qorvo, Greensboro, NC — May 2025 - Aug. 2025

- Developed heuristic algorithm compromising between greedy search and exhaustive search to find the best settings in an exponentially large discrete search space
- Designed algorithm to accept different objective functions to be adaptive to goals of circuit design engineers
- Used algorithm to find a set of feasible candidate solutions based on different objectives to facilitate circuit design

Teaching Assistant (4 times) Intro. Computer Systems, NC State University, Raleigh, NC — 2023 - 24